SGALAB

Evolution with the world

Global Training Scheme

Semi-active Suspension with GA

1200 - 1500
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Time table

- à 1200 – 1245 ***SGALAB***
  - ***What is GA***
  - ***SGALAB Quick Start***
- à 1245 – 1300 tea break
- à 1300 – 1345 ***Semi-active Suspension*** ***&*** ***Simulation***
- à 1345 – 1400 tea break
- à 1400 – 1430 ***Semi-active Suspension and GA***
- à 1430 – 1500 ***Q&A***

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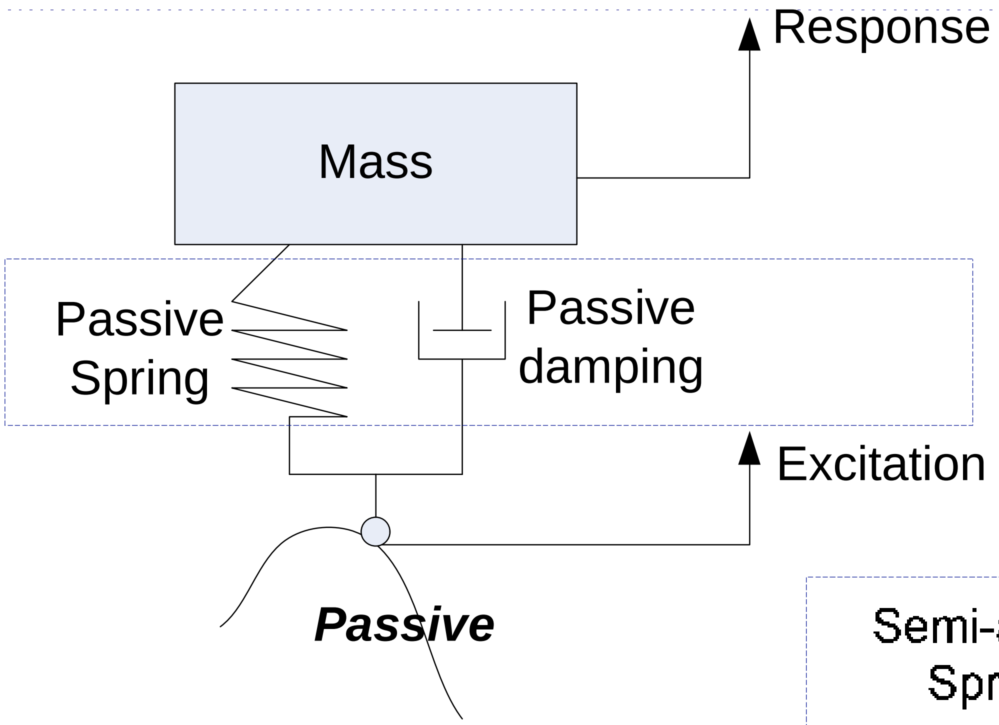
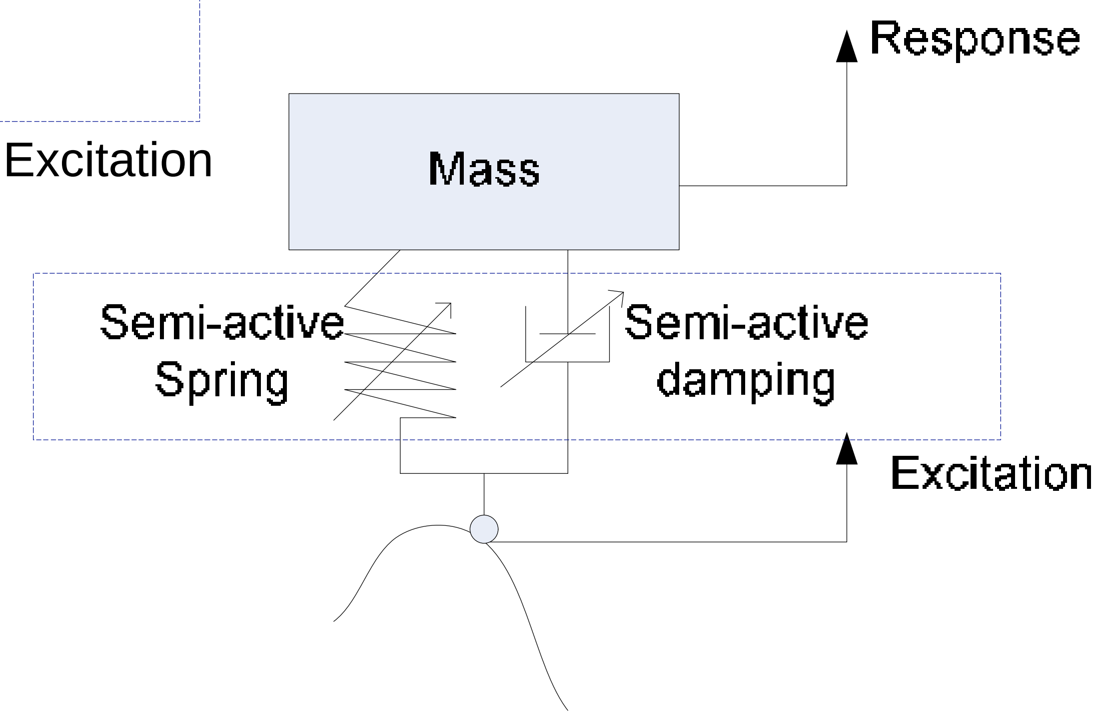
- Semi-active suspension introduction
- Semi-active suspension simulation
- Semi-active suspension paper list
- Objectives for the semi-active suspension
- Semi-active suspension fitness function design in SGALAB
- FAQ

Semi-active suspension definition

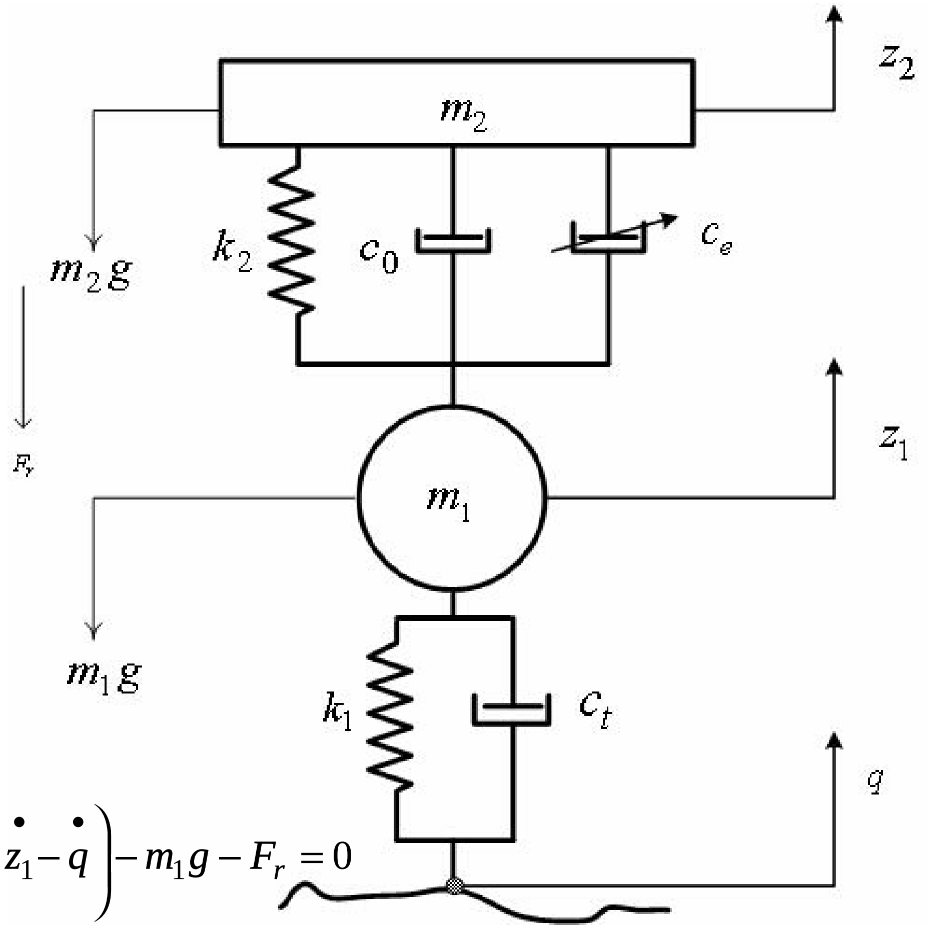


Passive



Semi-active

Semi-active suspension - 2-DOFs



$$\begin{cases} m_1 \ddot{z}_1 + (c_0 + c_e)\left(\dot{z}_1 - \dot{z}_2\right) + k_2(z_1 - z_2) + k_1(z_1 - q) + c_t\left(\dot{z}_1 - \dot{q}\right) - m_1 g - F_r = 0 \\ m_2 \ddot{z}_2 + (c_0 + c_e)\left(\dot{z}_2 - \dot{z}_1\right) + k_2(z_2 - z_1) - m_2 g = 0 \end{cases}$$

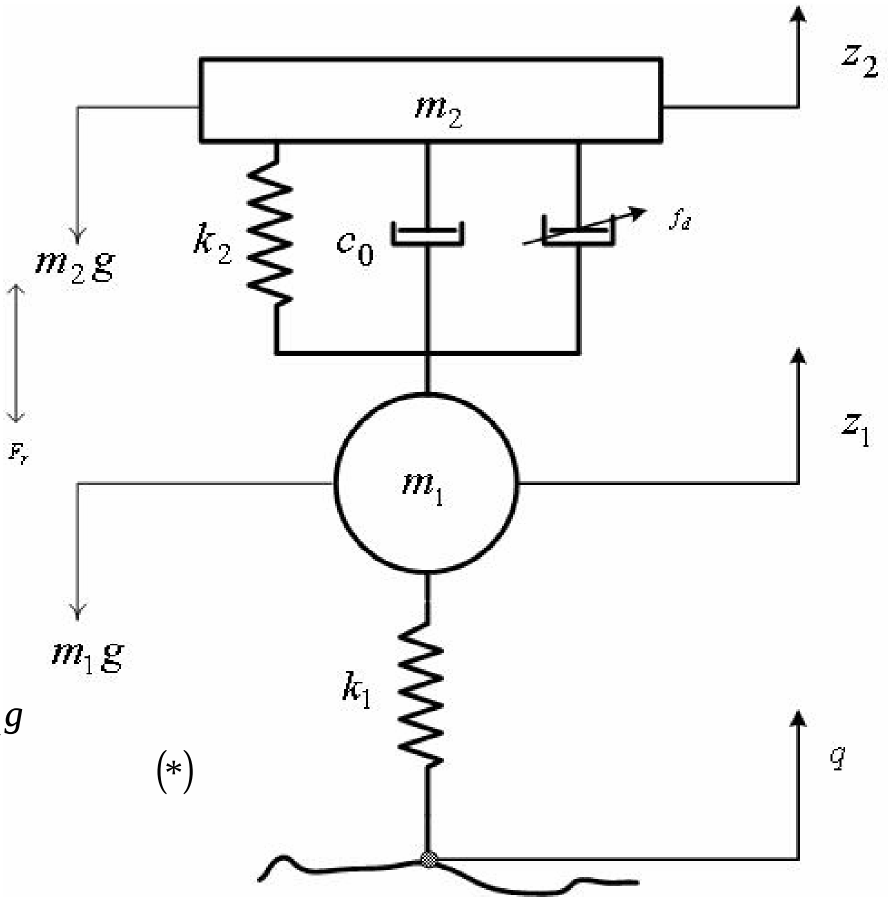
Semi-active suspension - 2-DOFs

to Let $c_e\left(\dot{z}_2-\dot{z}_1\right)=f_d$

and to Ignore $c_t\left(\dot{z}_1-\dot{q}\right)=F_t$

then:

$$\begin{cases} m_1\ddot{z}_1=-c_0\left(\dot{z}_1-\dot{z}_2\right)-k_2\left(z_1-z_2\right)-k_1\left(z_1-q\right)+f_d-F_r+m_1 g \\ m_2\ddot{z}_2=-c_0\left(\dot{z}_2-\dot{z}_1\right)-k_2\left(z_2-z_1\right)-f_d+F_r+m_2 g=0 \end{cases} \quad (*)$$



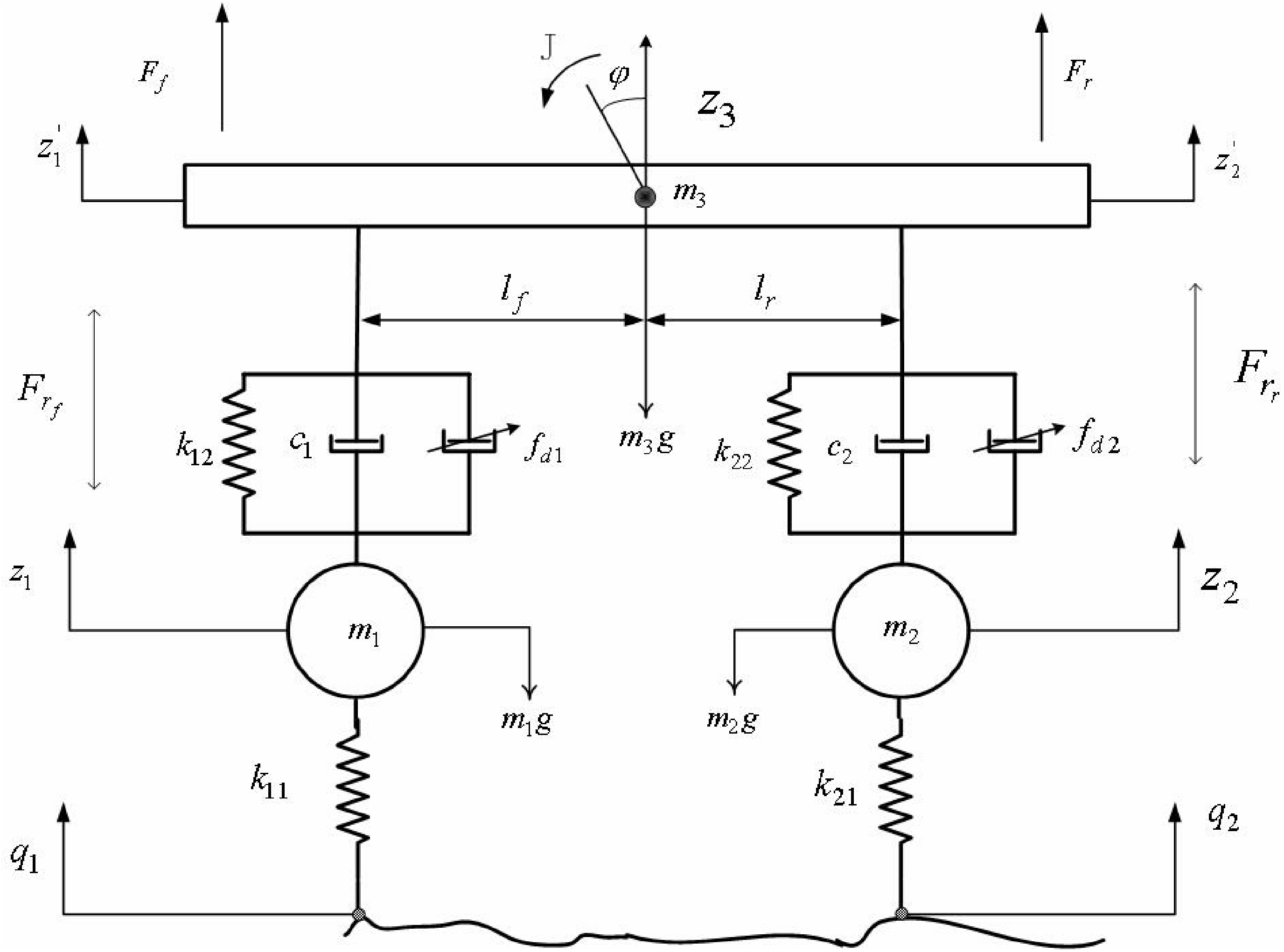
Semi-active suspension - 2-DOFs

State space: $\begin{cases} \dot{X} = AX + BQ + EU \\ Y = CX + DQ + FU \end{cases}$ where: $X = \begin{Bmatrix} z_1 - q \\ z_2 - z_1 \\ \dot{z}_1 \\ \dot{z}_2 \end{Bmatrix}$ $Y = \begin{Bmatrix} \ddot{z}_2 \\ z_1 - q \\ z_2 - z_1 \end{Bmatrix}$ $Q = \begin{Bmatrix} \dot{q} \\ g \\ F_r \end{Bmatrix}$

$$U = \{f_d\} \qquad A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 1 \\ \frac{-k_1}{m_1} & \frac{k_2}{m_1} & \frac{-c_0}{m_1} & \frac{c_0}{m_1} \\ 0 & \frac{-k_2}{m_2} & \frac{c_0}{m_2} & \frac{-c_0}{m_2} \end{bmatrix} \qquad B = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & \frac{-1}{m_1} \\ 0 & 1 & \frac{1}{m_2} \end{bmatrix}$$

$$C = \begin{bmatrix} 0 & \frac{-k_2}{m_2} & \frac{c_0}{m_2} & \frac{-c_0}{m_2} \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \qquad E = \begin{bmatrix} 0 \\ 0 \\ \frac{1}{m_1} \\ \frac{-1}{m_2} \end{bmatrix} \qquad F = \begin{bmatrix} \frac{-1}{m_2} \\ 0 \\ 0 \end{bmatrix} \qquad D = \begin{bmatrix} 0 & 1 & \frac{1}{m_2} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Semi-active suspension - 4-DOFs - Pitch



Semi-active suspension - 4-DOFs - Pitch

$$
\begin{cases}
m_1 \ddot{z}_1 = k_{11}(z_1 - q_1) + k_{12}\left(z_1' - z_1\right) + c_1\left(\dot{z}_1' - \dot{z}_1\right) + F_{r_f} + f_{d1} + m_1 g \\
m_2 \ddot{z}_2 = k_{21}(z_2 - q_2) + k_{22}\left(z_2' - z_2\right) + c_2\left(\dot{z}_2' - \dot{z}_2\right) + F_{r_r} + f_{d2} + m_2 g \\
m_3 \ddot{z}_3 = k_{12}\left(z_1 - z_1'\right) + k_{22}\left(z_2 - z_2'\right) + c_1\left(\dot{z}_1 - \dot{z}_1'\right) + c_2\left(\dot{z}_2 - \dot{z}_2'\right) - f_{d1} - f_{d2} - F_{r_f} - F_{r_r} + m_3 g \\
J \ddot{j} = -\left[k_{12}\left(z_1 - z_1'\right) + c_1\left(\dot{z}_1 - \dot{z}_1'\right)\right] l_f + \left[k_{22}\left(z_2 - z_2'\right) + c_2\left(\dot{z}_2 - \dot{z}_2'\right)\right] l_r - l_f f_{d1} + l_f F_{r_f} + l_r f_{d2} - l_r F_{r_r}
\end{cases}
$$

where

$$z_1' = z_3 - j\, l_f$$

$$z_2' = z_3 + j\, l_r$$

Semi-active suspension - 4-DOFs - Pitch

$$\begin{cases} \dot{X} = AX + BQ + EU \\ Y = CX + DQ + FU \end{cases}$$

where

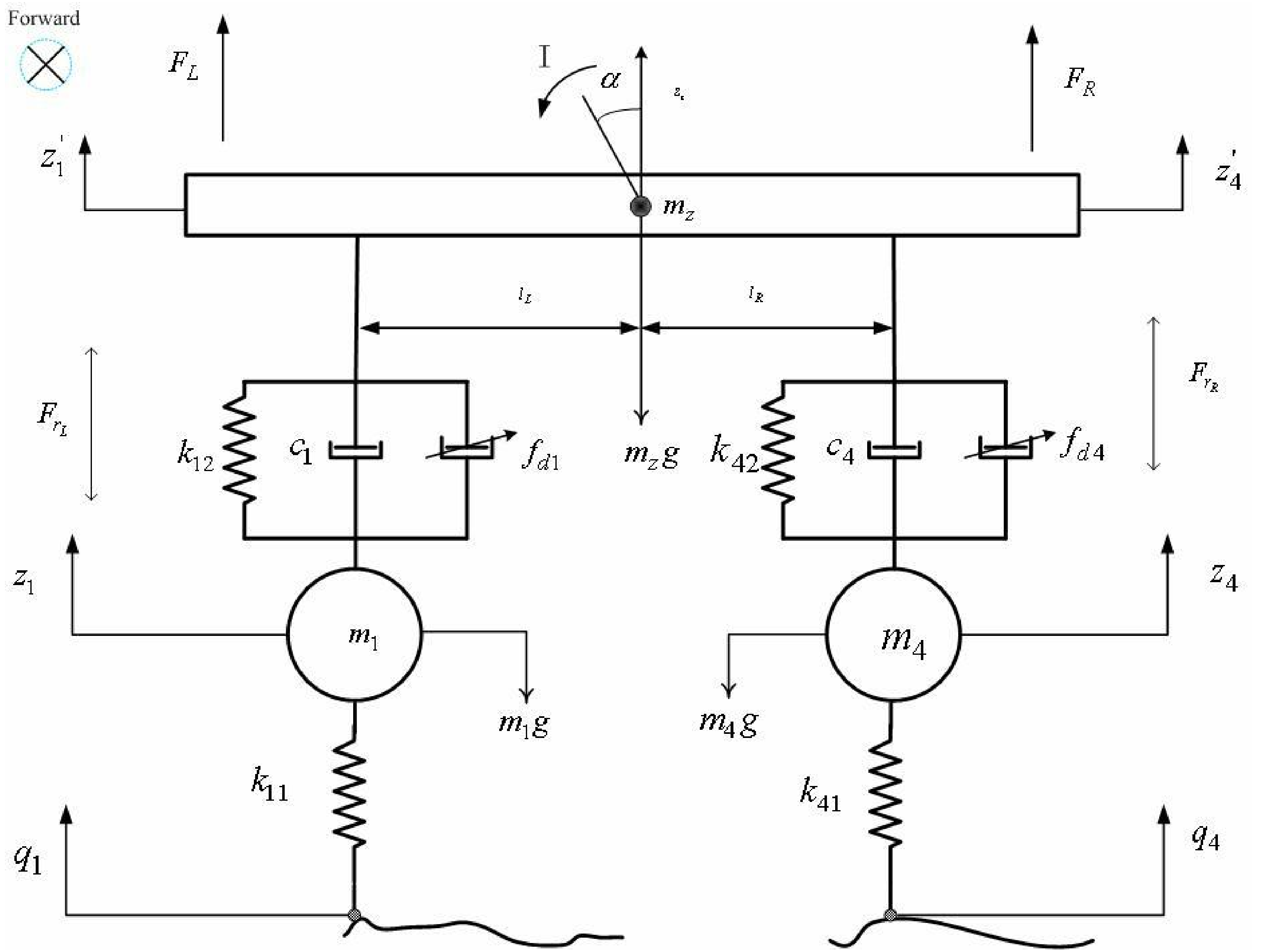
$$X = \begin{Bmatrix} z_1 - z_1' \\ z_2 - z_2' \\ q_1 - z_1 \\ q_2 - z_2 \\ \dot{z}_1 \\ \dot{z}_2 \\ \dot{z}_3 \\ \dot{j} \end{Bmatrix} \quad Y = \begin{Bmatrix} \ddot{z}_3 \\ z_1 - z_1' \\ z_2 - z_2' \\ q_1 - z_1 \\ q_2 - z_2 \end{Bmatrix} \quad U = \begin{Bmatrix} f_{d1} \\ f_{d2} \end{Bmatrix} \quad Q = \begin{Bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ g \\ F_{r_f} \\ F_{r_r} \end{Bmatrix} \quad D = \begin{bmatrix} 0 & 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$A = \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 & -1 & l_f \\ 0 & 0 & 0 & 0 & 0 & 1 & -1 & -l_r \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ \frac{-k_{12}}{m_1} & 0 & \frac{-k_{11}}{m_1} & 0 & \frac{-c_1}{m_1} & 0 & \frac{c_1}{m_1} & \frac{-c_1 l_f}{m_1} \\ 0 & \frac{-k_{22}}{m_2} & 0 & \frac{-k_{21}}{m_2} & 0 & \frac{-c_2}{m_2} & \frac{c_2}{m_2} & \frac{c_2}{m_2} \\ \frac{k_{12}}{m_3} & \frac{k_{22}}{m_3} & 0 & 0 & \frac{c_1}{m_3} & \frac{c_2}{m_3} & \frac{-(c_1+c_2)}{m_3} & \frac{l_f c_1 - l_r c_2}{m_3} \\ -\frac{k_{12} l_f}{J} & \frac{k_{22} l_r}{J} & 0 & 0 & \frac{-c_1 l_f}{J} & \frac{c_2 l_r}{J} & \frac{c_1 l_f - c_2 l_r}{J} & \frac{-\left(c_1 l_f^2 + c_2 l_r^2\right)}{J} \end{bmatrix} \quad B = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & l_f & -l_r \end{bmatrix}$$

Semi-active suspension - 4-DOFs - Pitch

$$E=\begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ \frac{1}{m_1} & 0 \\ 0 & \frac{1}{m_2} \\ -\frac{1}{m_3} & -\frac{1}{m_3} \\ \frac{-l_f}{J} & \frac{l_r}{J} \end{bmatrix}
\quad
C=\begin{bmatrix} \frac{k_{12}}{m_3} & \frac{k_{22}}{m_3} & 0 & 0 & \frac{c_1}{m_3} & \frac{c_2}{m_3} & \frac{-(c_1+c_2)}{m_3} & \frac{l_f c_1 - l_r c_2}{m_3} \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{bmatrix}
\quad
F=\begin{bmatrix} \frac{1}{m_3} & \frac{1}{m_3} \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Semi-active suspension - 4-DOFs - Roll



Semi-active suspension - 4-DOFs - Roll

$$
\begin{cases}
m_1 \ddot{z}_1 = k_{11}(z_1 - q_1) + k_{12}(z_1' - z_1) + c_1\left(\dot{z}_1' - \dot{z}_1\right) + F_{r_L} + f_{d1} + m_1 g \\
m_4 \ddot{z}_4 = k_{41}(z_4 - q_4) + k_{42}(z_4' - z_4) + c_4\left(\dot{z}_4' - \dot{z}_4\right) + F_{r_R} + f_{d4} + m_4 g \\
m_z \ddot{z}_z = k_{12}(z_1 - z_1') + k_{42}(z_4 - z_4') + c_1\left(\dot{z}_1 - \dot{z}_1'\right) + c_4\left(\dot{z}_4 - \dot{z}_4'\right) - f_{d1} - f_{d4} - F_{r_L} - F_{r_R} + m_z g \\
I \ddot{a} = -\left[k_{12}(z_1 - z_1') + c_1\left(\dot{z}_1 - \dot{z}_1'\right)\right] l_L + \left[k_{42}(z_4 - z_4') + c_4\left(\dot{z}_4 - \dot{z}_4'\right)\right] l_R - l_L f_{d1} + l_L F_{r_L} + l_R f_{d4} - l_R F_{r_R}
\end{cases}
$$

where $z_1' = z_z - a l_L$

$z_4' = z_z + a l_R$

Semi-active suspension - 4-DOFs - Roll

$$\begin{cases} \dot{X} = AX + BQ + EU \\ Y = CX + DQ + FU \end{cases}$$

$$X = \begin{Bmatrix} z_1 - z_1' \\ z_4 - z_4' \\ q_1 - z_1 \\ q_4 - z_4 \\ \dot{z}_1 \\ \dot{z}_4 \\ \dot{z}_z \\ \dot{a} \end{Bmatrix} \quad Y = \begin{Bmatrix} \ddot{z}_z \\ z_1 - z_1' \\ z_4 - z_4' \\ q_1 - z_1 \\ q_4 - z_4 \end{Bmatrix} \quad U = \begin{Bmatrix} f_{d1} \\ f_{d4} \end{Bmatrix} \quad Q = \begin{Bmatrix} \dot{q}_1 \\ \dot{q}_4 \\ g \\ F_{r_L} \\ F_{r_R} \end{Bmatrix}$$

$$B = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & l_L & -l_R \end{bmatrix}$$

$$A = \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 & -1 & l_f \\ 0 & 0 & 0 & 0 & 0 & 1 & -1 & -l_r \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ \frac{-k_{12}}{m_1} & 0 & \frac{-k_{11}}{m_1} & 0 & \frac{-c_1}{m_1} & 0 & \frac{c_1}{m_1} & \frac{-c_1 l_f}{m_1} \\ 0 & \frac{-k_{42}}{m_4} & 0 & \frac{-k_{41}}{m_4} & 0 & \frac{-c_4}{m_4} & \frac{c_4}{m_4} & \frac{c_4}{m_4} \\ \frac{k_{12}}{m_z} & \frac{k_{42}}{m_z} & 0 & 0 & \frac{c_1}{m_z} & \frac{c_4}{m_z} & \frac{-(c_1 + c_4)}{m_z} & \frac{l_L c_1 - l_R c_4}{m_z} \\ -\frac{k_{12} l_L}{I} & \frac{k_{42} l_R}{I} & 0 & 0 & \frac{-c_1 l_L}{I} & \frac{c_4 l_R}{I} & \frac{c_1 l_L - c_4 l_R}{I} & \frac{-\left(c_1 l_L^2 + c_4 l_R^2\right)}{I} \end{bmatrix}$$

Semi-active suspension - 4-DOFs - Roll

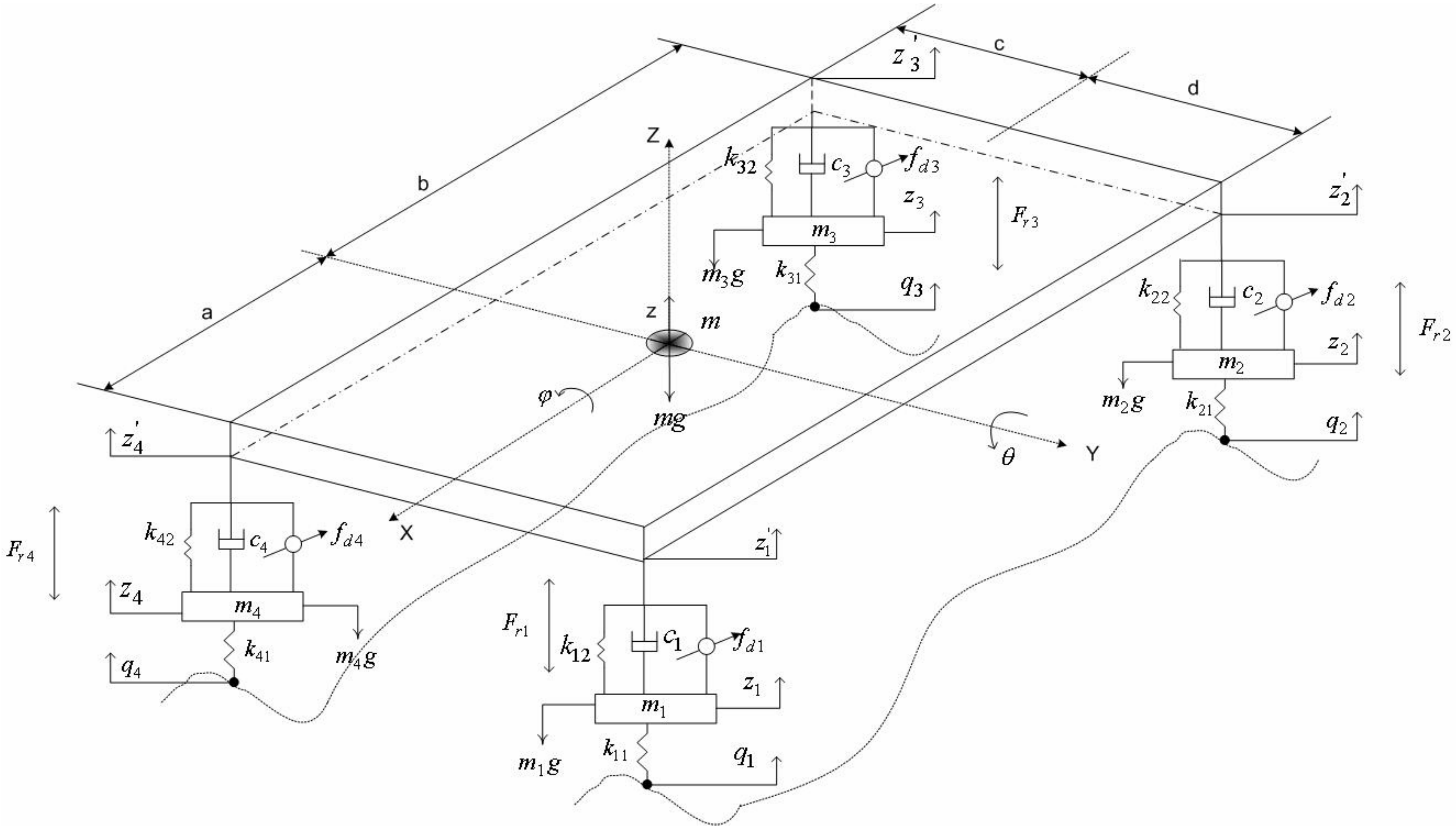
$$D = \begin{bmatrix} 0 & 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$E = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ \frac{1}{m_1} & 0 \\ 0 & \frac{1}{m_4} \\ -\frac{1}{m_z} & -\frac{1}{m_z} \\ \frac{-l_L}{I} & \frac{l_R}{I} \end{bmatrix}$$

$$F = \begin{bmatrix} \frac{1}{m_z} & \frac{1}{m_z} \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$C = \begin{bmatrix} \frac{k_{12}}{m_z} & \frac{k_{42}}{m_z} & 0 & 0 & \frac{c_1}{m_z} & \frac{c_4}{m_z} & \frac{-(c_1 + c_4)}{m_z} & \frac{l_L c_1 - l_R c_4}{m_z} \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Semi-active suspension - 7-DOFs



Semi-active suspension - 7-DOFs

$$
\begin{cases}
m_1 \ddot{z}_1 = k_{11}(q_1 - z_1) + k_{12}\left(z_1' - z_1\right) + c_1\left(\dot{z}_1' - \dot{z}_1\right) + f_{d1} + F_{r1} + m_1 g \\
m_2 \ddot{z}_2 = k_{21}(q_2 - z_2) + k_{22}\left(z_2' - z_2\right) + c_2\left(\dot{z}_2' - \dot{z}_2\right) + f_{d2} + F_{r2} + m_2 g \\
m_3 \ddot{z}_3 = k_{31}(q_3 - z_3) + k_{32}\left(z_3' - z_3\right) + c_3\left(\dot{z}_3' - \dot{z}_3\right) + f_{d3} + F_{r3} + m_3 g \\
m_4 \ddot{z}_4 = k_{41}(q_4 - z_4) + k_{42}\left(z_4' - z_4\right) + c_4\left(\dot{z}_4' - \dot{z}_4\right) + f_{d4} + F_{r4} + m_4 g \\
m \ddot{z} = k_{12}\left(z_1 - z_1'\right) + k_{22}\left(z_2 - z_2'\right) + k_{32}\left(z_3 - z_3'\right) + k_{42}\left(z_4 - z_4'\right) + c_1\left(\dot{z}_1 - \dot{z}_1'\right) + c_2\left(\dot{z}_2 - \dot{z}_2'\right) + c_3\left(\dot{z}_3 - \dot{z}_3'\right) + c_4\left(\dot{z}_4 - \dot{z}_4'\right) \\
- f_{d1} - f_{d2} - f_{d3} - f_{d4} + mg - F_{r1} - F_{r2} - F_{r3} - F_{r4} \\
J_x \ddot{j} = -\left[k_{32}\left(z_3 - z_3'\right) + c_3\left(\dot{z}_3 - \dot{z}_3'\right) + k_{42}\left(z_4 - z_4'\right) + c_4\left(\dot{z}_4 - \dot{z}_4'\right)\right]c + \left[k_{12}\left(z_1 - z_1'\right) + c_1\left(\dot{z}_1 - \dot{z}_1'\right) + k_{22}\left(z_2 - z_2'\right) + c_2\left(\dot{z}_2 - \dot{z}_2'\right)\right]d \\
-\left(f_{d3} + f_{d4}\right)c + \left(F_{r3} + F_{r4}\right)c + \left(f_{d1} + f_{d2}\right)d - \left(F_{r1} + F_{r2}\right)d \\
J_y \ddot{q} = -\left[k_{12}\left(z_1 - z_1'\right) + c_1\left(\dot{z}_1 - \dot{z}_1'\right) + k_{42}\left(z_4 - z_4'\right) + c_4\left(\dot{z}_4 - \dot{z}_4'\right)\right]a + \left[k_{22}\left(z_2 - z_2'\right) + c_2\left(\dot{z}_2 - \dot{z}_2'\right) + k_{32}\left(z_3 - z_3'\right) + c_3\left(\dot{z}_3 - \dot{z}_3'\right)\right]b \\
-\left(f_{d1} + f_{d4}\right)a + \left(F_{r1} + F_{r4}\right)a + \left(f_{d2} + f_{d3}\right)b - \left(F_{r2} + F_{r3}\right)b
\end{cases}
$$

where

$$
\begin{aligned}
z_1' &= z - \left(aq + dj\right) \\
z_2' &= z + \left(bq - dj\right) \\
z_3' &= z + \left(bq + cj\right) \\
z_4' &= z + \left(aq - cj\right)
\end{aligned}
$$

Semi-active suspension - 7-DOFs

$$\begin{cases} \dot{X} = AX + BQ + EU \\ Y = CX + DQ + FU \end{cases}$$ where

$$U = \begin{Bmatrix} f_{d1} \\ f_{d2} \\ f_{d3} \\ f_{d4} \end{Bmatrix} \quad Q = \begin{Bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \\ \dot{q}_4 \\ g \\ F_{r1} \\ F_{r2} \\ F_{r3} \\ F_{r4} \end{Bmatrix} \quad X = \begin{Bmatrix} z_1 - z_1' \\ z_2 - z_2' \\ z_3 - z_3' \\ z_4 - z_4' \\ q_1 - z_1 \\ q_2 - z_2 \\ q_3 - z_3 \\ q_4 - z_4 \\ \dot{z}_1 \\ \dot{z}_2 \\ \dot{z}_3 \\ \dot{z}_4 \\ \dot{z} \\ \dot{j} \\ \dot{q} \end{Bmatrix}$$

$$E = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \frac{1}{m_1} & 0 & 0 & 0 \\ 0 & \frac{1}{m_2} & 0 & 0 \\ 0 & 0 & \frac{1}{m_3} & 0 \\ 0 & 0 & 0 & \frac{1}{m_4} \\ -\frac{1}{m} & -\frac{1}{m} & -\frac{1}{m} & -\frac{1}{m} \\ \frac{d}{J_x} & \frac{d}{J_x} & \frac{-c}{J_x} & \frac{-c}{J_x} \\ \frac{-a}{J_y} & \frac{b}{J_y} & \frac{b}{J_y} & \frac{-a}{J_y} \end{bmatrix} \quad Y = \begin{Bmatrix} \ddot{z} \\ \ddot{j} \\ \ddot{q} \\ z_1 - z_1' \\ z_2 - z_2' \\ z_3 - z_3' \\ z_4 - z_4' \\ q_1 - z_1 \\ q_2 - z_2 \\ q_3 - z_3 \\ q_4 - z_4 \end{Bmatrix}$$

Semi-active suspension - 7-DOFs

$$
A=\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & -1 & d & a \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & -1 & d & -b \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & -1 & -c & -b \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & -1 & c & -a \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\
\frac{-k_{12}}{m_1} & 0 & 0 & 0 & \frac{k_{11}}{m_1} & 0 & 0 & 0 & \frac{-c_1}{m_1} & 0 & 0 & 0 & \frac{c_1}{m_1} & \frac{-c_1 d}{m_1} & \frac{-c_1 a}{m_1} \\
0 & \frac{-k_{22}}{m_2} & 0 & 0 & 0 & \frac{k_{21}}{m_2} & 0 & 0 & 0 & \frac{-c_2}{m_2} & 0 & 0 & \frac{c_2}{m_2} & \frac{-c_2 d}{m_2} & \frac{c_2 b}{m_2} \\
0 & 0 & \frac{-k_{32}}{m_3} & 0 & 0 & 0 & \frac{k_{31}}{m_3} & 0 & 0 & 0 & \frac{-c_3}{m_3} & 0 & \frac{c_3}{m_3} & \frac{c_3 c}{m_3} & \frac{c_3 b}{m_3} \\
0 & 0 & 0 & \frac{-k_{42}}{m_4} & 0 & 0 & 0 & \frac{k_{41}}{m_4} & 0 & 0 & 0 & \frac{-c_4}{m_4} & \frac{c_4}{m_4} & \frac{-c_4 c}{m_4} & \frac{c_4 a}{m_4} \\
\frac{k_{12}}{m} & \frac{k_{22}}{m} & \frac{k_{32}}{m} & \frac{k_{42}}{m} & 0 & 0 & 0 & 0 & \frac{c_1}{m} & \frac{c_2}{m} & \frac{c_3}{m} & \frac{c_4}{m} & \frac{-(c_1+c_2+c_3+c_4)}{m} & \frac{(c_1+c_2)d-(c_3-c_4)c}{m} & \frac{(c_1-c_4)a-(c_2+c_3)b}{m} \\
\frac{k_{12}d}{J_x} & \frac{k_{22}d}{J_x} & \frac{-k_{32}c}{J_x} & \frac{-k_{42}c}{J_x} & 0 & 0 & 0 & 0 & \frac{c_1 d}{J_x} & \frac{c_2 d}{J_x} & \frac{-c_3 c}{J_x} & \frac{c_4 c}{J_x} & \frac{(c_3-c_4)c-(c_1+c_2)d}{J_x} & \frac{(c_3+c_4)c^2+(c_1+c_2)d^2}{J_x} & \frac{(c_3 b-c_4 a)c+(c_1 a-c_2 b)d}{J_x} \\
\frac{-k_{12}a}{J_y} & \frac{k_{22}b}{J_y} & \frac{k_{32}b}{J_y} & \frac{k_{42}a}{J_y} & 0 & 0 & 0 & 0 & \frac{c_1 a}{J_y} & \frac{c_2 b}{J_y} & \frac{c_3 b}{J_y} & \frac{c_4 a}{J_y} & -\frac{(c_1+c_4)a+(c_2+c_3)b}{J_y} & \frac{(c_1 a+c_2 b)d+(c_4 a-c_3 b)c}{J_y} & \frac{(c_1-c_4)a^2-(c_2+c_3)b^2}{J_y}
\end{bmatrix}
$$

Semi-active suspension - 7-DOFs

F_{r_1}

is a constant friction for 1 - front left half suspension

F_{r_2}

is a constant friction for 2 - rear left half suspension

F_{r_3}

is a constant friction for 3 - rear right half suspension

F_{r_4}

is a constant friction for 4 - front right half suspension

$$B = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & -1 & -1 & -1 & -1 \\ 0 & 0 & 0 & 0 & 0 & \frac{-d}{J_x} & \frac{-d}{J_x} & \frac{c}{J_x} & \frac{c}{J_x} \\ 0 & 0 & 0 & 0 & 0 & \frac{-a}{J_y} & \frac{b}{J_y} & \frac{b}{J_y} & \frac{-a}{J_y} \end{bmatrix}$$

Semi-active suspension - 7-DOFs

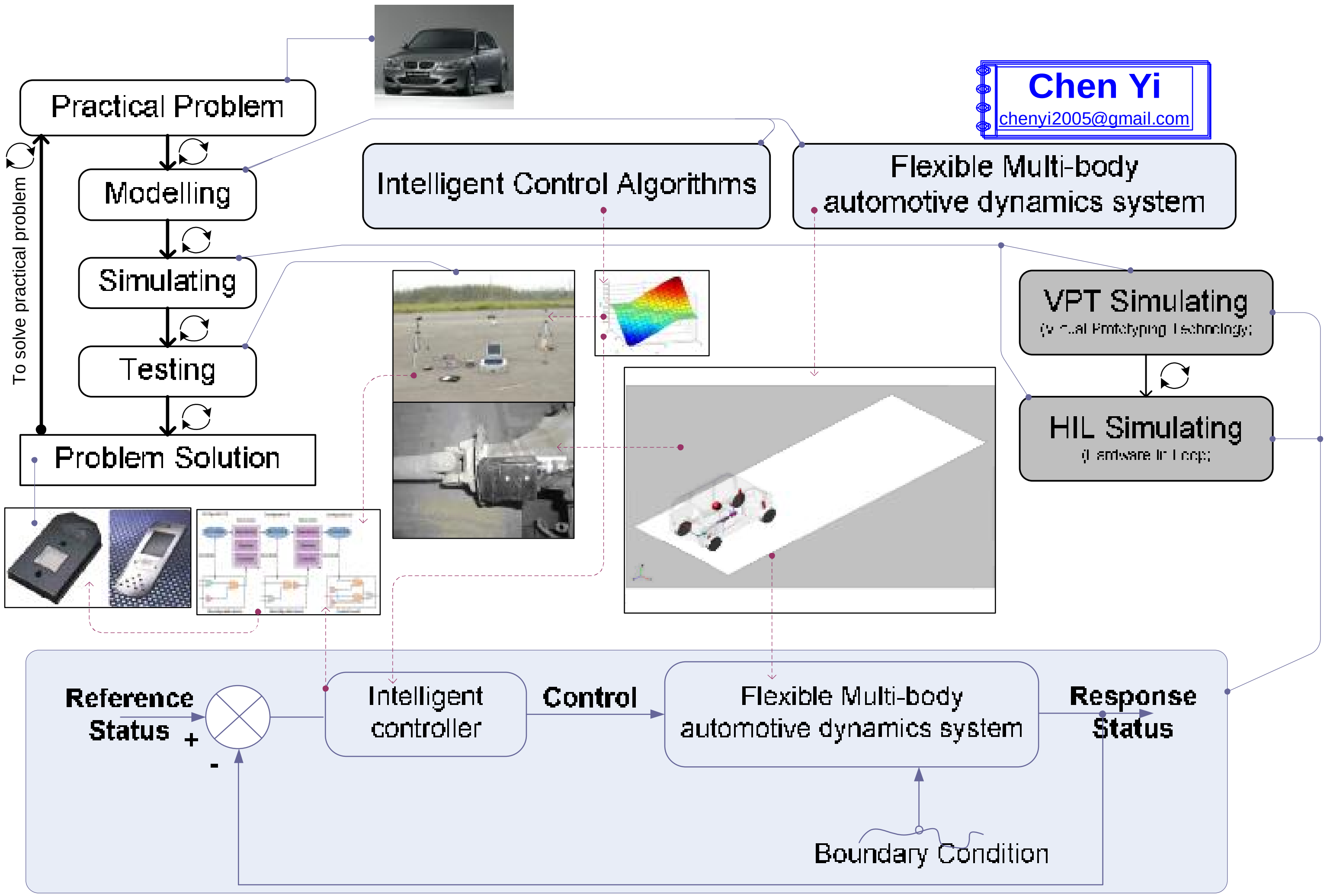
$$
C = \begin{bmatrix}
\frac{k_{12}}{m} & \frac{k_{22}}{m} & \frac{k_{32}}{m} & \frac{k_{42}}{m} & 0 & 0 & 0 & 0 & \frac{c_1}{m} & \frac{c_2}{m} & \frac{c_3}{m} & \frac{c_4}{m} & \frac{-(c_1+c_2+c_3+c_4)}{m} & \frac{(c_1+c_2)d-(c_3-c_4)c}{m} & \frac{(c_1-c_4)a-(c_2+c_3)b}{m} \\
\frac{k_{12}d}{J_x} & \frac{k_{22}d}{J_x} & \frac{-k_{32}c}{J_x} & \frac{-k_{42}c}{J_x} & 0 & 0 & 0 & 0 & \frac{c_1 d}{J_x} & \frac{c_2 d}{J_x} & \frac{-c_3 c}{J_x} & \frac{c_4 c}{J_x} & \frac{(c_3-c_4)c-(c_1+c_2)d}{J_x} & \frac{(c_3+c_4)c^2+(c_1+c_2)d^2}{J_x} & \frac{(c_3 b-c_4 a)c+(c_1 a-c_2 b)d}{J_x} \\
\frac{-k_{12}a}{J_y} & \frac{k_{22}b}{J_y} & \frac{k_{32}b}{J_y} & \frac{k_{42}a}{J_y} & 0 & 0 & 0 & 0 & \frac{c_1 a}{J_y} & \frac{c_2 b}{J_y} & \frac{c_3 b}{J_y} & \frac{c_4 a}{J_y} & -\frac{(c_1+c_4)a+(c_2+c_3)b}{J_y} & \frac{(c_1 a+c_2 b)d+(c_4 a-c_3 b)c}{J_y} & \frac{(c_1-c_4)a^2-(c_2+c_3)b^2}{J_y} \\
1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}
$$

Semi-active suspension - 7-DOFs

$$D=\begin{bmatrix} 0 & 0 & 0 & 0 & 1 & -1 & -1 & -1 & -1 \\ 0 & 0 & 0 & 0 & 0 & \frac{-d}{J_x} & \frac{-d}{J_x} & \frac{c}{J_x} & \frac{c}{J_x} \\ 0 & 0 & 0 & 0 & 0 & \frac{-a}{J_x} & \frac{b}{J_x} & \frac{b}{J_x} & \frac{-a}{J_x} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad F=\begin{bmatrix} -\frac{1}{m} & -\frac{1}{m} & -\frac{1}{m} & -\frac{1}{m} \\ \frac{d}{J_x} & \frac{d}{J_x} & \frac{-c}{J_x} & \frac{-c}{J_x} \\ \frac{-a}{J_y} & \frac{b}{J_y} & \frac{b}{J_y} & \frac{-a}{J_y} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Contents

Semi-active suspension simulation outline



Simulation case

- 1) SGA__suspension_skyhook_quarter.mdl
- 2) SGA__suspension_flc_quarter.mdl

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- FAQ

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Contents

Objectives for the semi-active suspension

The factor ***J*** is defined as the control objective of semi-active suspension:

$$J = \sqrt{\frac{J_1^2 \times w_1 + J_2^2 \times w_2 + J_3^2 \times w_3}{w_1 + w_2 + w_3}}$$

where,

J_1 is the is **ITAE** factor of vertical accelerator ;

J_2 is the **ITAE** factor of suspensions flexibility;

J_3 is the **ITAE** factor of tyre dynamic loads .

ω_1, ω_2 , ω_3 are the weights of factors

ITAE - Integral of Time-multiplied Absolute value of Error

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Semi-active suspension fitness function design in SGALAB

- A fitness function must be devised for each problem to be solved. Given a particular chromosome, a solution, the fitness function returns a single numerical fitness which is supposed to be proportional to the utility or adaptation of the individual which that chromosome represents. A simple fitness function for semi-active suspension that reflects small steady-state errors, a short rise-time, low oscillations and overshoots with a good stability is given by :

$$Fitness = \frac{1}{J + e}$$

- Where, ε is the parameter to avoid Fitness = ∞, ε = 0.2 , and J is ITAE evaluation factor and it is a objective function of system output or system output error , with higher values corresponding to better controller performance.

- A fitness function example in sgalab

FAQ – list

- How GA can searching the optimal result, producing the Pareto Optimal curve and furthermore how GA do the ranking?

  Show GA workflow in SGALAB_FAQ_QuickStart

- Which parts in the programs need to be changed or considered if (a) I want to add or remove the objective function, constraint and side constraint (b) I want to combine, link or integrate the problems in C above?

  Show GA workflow in SGALAB_FAQ_QuickStart

- How GA processing all the data?

  Show GA workflow in SGALAB_FAQ_QuickStart

- How to encoding and coding the kinematic and ride problem ?

  Show GA workflow in SGALAB_FAQ_QuickStart

FAQ – list

- What is the meaning of 'niching'?

  FAQ - 1

- Is there any mathematical involve in GA beside of objective function, constraint and side constraint ?

  yes, see SGALAB_FAQ_QuickStart

- How to verify the GA result ?

  FAQ - 2

- Can you make me to have FULLY and DEEP understand on the GA process without I know the program language.   For example how GA do rank, elitism, distribution, min max , weighted or so on? I'm not sure about this.

  Show GA workflow in SGALAB_FAQ_QuickStart

- What is the different between SGALAB with MOGA for example ? what is do you mean by "simple" in front of SGALAB?

  See SGALAB_FAQ_QuickStart

FAQ – 1 what is Niching?

In ecology, a **niche** (pronounced nich, neesh or nish) is a term describing the relational position of a species or population in its ecosystem.

The ecological niche describes how an organism or population responds to the distribution of resources and competitors (e. g., by growing when resources are abundant, and predators, parasites and pathogens are scarce) and how it in turn alters those same factors (e.g., limiting access to resources by other organisms, acting as a food source for predators and a consumer of prey).

FAQ – 1 what is Niching?

- **Niching** is a technique based on the metaphor from nature that different species or subpopulations have different niches. There is no competition between niches, however within a niche there is competition.
- Within GAS, niching allows different parts of the fitness function to be explored in parallel, with convergence to several rather than just one function maximum.

FAQ – 1 what is Niching?

Goldberg and Richardson asked and answered these two niching questions in paper [6]:

Who should share?

How much should be shared?

The essence of the answers to these questions is to consider the degree of similarity between two individuals in the GA population.

$$m_i = \sum_{j=1}^{n} sh(d_{ij})$$

$$sh(d_{ij}) = \begin{cases} 1 - \left(\dfrac{d_{ij}}{\sigma_{sh}} \right)^{\alpha} & if \quad d_{ij} < \sigma_{sh} \\ 0 & otherwise \end{cases}$$

$$\sigma_{sh} = \frac{\sqrt{k}}{2 * \sqrt[k]{m}}$$

Niching Papers

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FAQ - 2 How to verify the GA results ?

- 1) Practical data to verify
- 2) Large population or scale calculation
- 3) Statistics methods

FAQ - 3 Zoom in MO methods in SGALAB

- 1) VEGA
- 2) MOGA
- 2) NPGA
- 3) NSGA
- 4) NSGAII

SGALAB

Evolution with the world

Global Training Scheme

Semi-active Suspension with GA

1200 -1500
29-Dec-2007

Leo Chen
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End